



SpendR

presented by: **MARCO POLO**



Github Repository

SpendR- AI-Powered Cloud
Cost Optimization Platform

Click on the link above

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The Problem

Every company that uses cloud servers, startups, hospitals, banks, universities pays Amazon or Google by the hour for those servers whether they're being used or not.

Studies show that around 35% of cloud spend is wasted on idle or oversized resources that nobody noticed. Globally that's over \$26 billion wasted every year.

The problem isn't that companies are careless it's that there's no intelligent system watching in real time and acting automatically. By the time someone notices, the bill has already arrived.



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Proposed solution

An autonomous cloud cost intelligence system that:

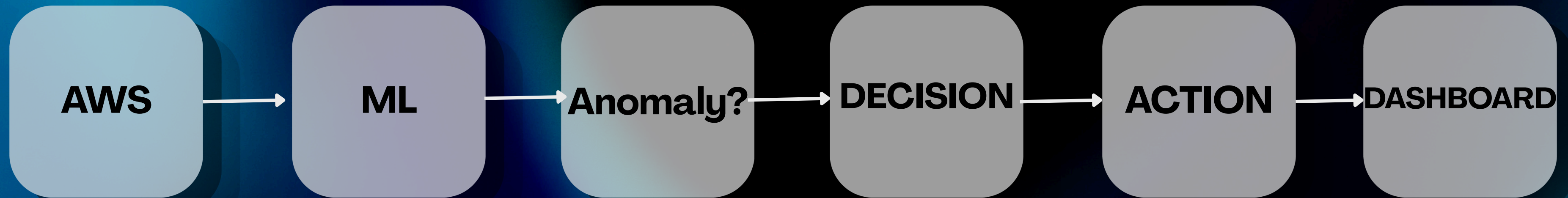
- Monitors AWS servers and their costs in real time
- Detects wasteful or abnormal spending using machine learning
- Automatically fixes the problem — no human intervention required
- Displays the savings live on a real-time dashboard

“ Problem statement requirements

- Provision of real cloud resources
- Build cost telemetry pipeline
- ML anomaly detection (Isolation forest/Prophet)
- Automatic optimization via APIs given AWS SDK (Boto 3)
- Real time dashboard with cost graphs , alerts and actions taken

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“System Architecture



ML model Backend

Model used: Isolation Forest

Learns normal behaviour and detects anomalies

- Training phase
- Detection phase
- Decision logic
- Confidence based action

Added key features:

- Savings Impact Engine
- Confidence-Based Automation
- Smart Alerts
- Multi-Resource Intelligence
- Predictive Cost Forecast
- Safe Mode
- Simulation Mode
- Recommendation Engine
- Logging System

Dashboard Frontend

Flask + HTML dashboard

Provides a clear, real-time view of cloud usage, anomalies, and cost optimization actions.

- Live Metrics Panel
- Automated Actions Display
- Visualization Graphs
- Anomaly Detection Alerts
- Cost & Savings Tracker



Integration

Connected directly with backend system to reflect real-time data, anomaly detection, and automated actions.

Cloud Cost Intelligence

Live

Total cost today

\$4.82

Anomalies detected

3

Saved so far

\$1.20

Cost over time



Recent actions

ec2-instance-idle-01

auto stopped · saved \$0.60

lambda-runaway-fn

concurrency limited · saved \$0.40

s3-unused-volume-09

reviewing...

“Team Contributions

Rida Azam

AWS Cloud Setup

Shreya L

ML (Backend)

Niharika G

Dashboard (Frontend)

Avani Verma

Research, Design & Integration



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Thank You.